

CONTACT	6 Snowdrop Close, Loughborough, UK, LE11 2WG	Mobile: +447540983126 E-mail: gerard.mccaul@gmail.com Google Scholar gmccaul.co.uk
RESEARCH PROFILE	I am a theoretical physicist and writer. My research covers a broad array of topics, including quantum control, dynamical systems theory, and neuromorphic computing. At present my principal efforts concern the development of a thermodynamic framework that derives fundamental limits on computation, learning, and physical representation. The goal is architecture-agnostic efficiency bounds for representation-bearing systems that are analytically sharp and empirically testable on current hardware, with a particular focus on reservoir and neuromorphic computing. A parallel project translates this framework into public writing, on the view that the reach of a foundational theory depends partly on how many people can reason with it.	
EMPLOYMENT	Loughborough University	2024–Present
	PDRA, QRC-4-ESP project	
	Tulane University, New Orleans	2019–2024
	- W.M. Keck Fellow - Postdoctoral Fellow	2021–2024 2019–2021
	Qidong Middle School, Jiangsu, China	2013–2014
	IB and IGCSE mathematics and physics teacher	
EDUCATION	King's College London	2015–2019
	- PhD, Physics — <i>Stochastic Representations of Open Systems</i> — awarded March 2019 - Supervisors: Lev Kantorovich, Chris Lorenz	
	King's College London	2014–2015
	MRes, Non-Equilibrium Systems: Theoretical Modelling, Simulation and Data Analysis (Distinction)	
	University of Warwick	2009–2013
	MPhys, Physics (First Class)	
PUBLICATIONS	Corresponding Author: * Equal Contribution: †	
	“Mean-Force Hamiltonians from Influence Functionals” Gerard McCaul*, arXiv:2602.13146 (2026). “Observables in Motion: A Guide to Simulating Classical and Quantum Dynamics” Denys I. Bondar, Gerard McCaul, Andrii Sotnikov, arXiv:2509.20403 (2025). <i>A textbook-length treatment of Hilbert-space representations of dynamics; I contributed the introductory chapters, open-systems extensions, and quantum-classical mapping sections.</i> “Free Snacks in Quantum Complexity” Gerard McCaul*, arXiv:2509.04618 (2025). <i>A complexity-theoretic analysis of the ITQDE methodology: scaling and stability criteria, a principled bias-variance smoothing, and the identification of a regime where coarse-grained spectral information is obtainable with only polynomial resources.</i> “Unwrapping Photonic Reservoirs: Enhanced Expressivity via Random Fourier Encoding over Stretched Domains” Gerard McCaul, Girish Tripathy, Giulia Marcucci, Juan Sebastian Toterogongora*, <i>Chaos</i> 35, 093129 (2025) — Featured Article	

- “Bose–Einstein Condensate Sub-wavelength Confinement via Superoscillations”
Dusty R. Lindberg, **Gerard McCaul**, Peisong Peng, Lev Kaplan, Diyar Talbayev, Denys I. Bondar, *Low Temp. Phys.* **51**, 1471 (2025)
- “Tuning the Terahertz Response of Liquids by Creating Polar Many-body Excitations”
Gerard McCaul^{*,†}, Matthias Runge[†], Michael Woerner, Diyar Talbayev, Thomas Elsaesser, Denys I. Bondar, *Phys. Rev. A* **112**, L011101 (2025) — **Letter**, *Chemistry World* coverage
- “Minimal Quantum Reservoirs with Hamiltonian Encoding”
Gerard McCaul^{*}, Juan Sebastian Toterogongora, Wendy Otieno, Sergey Savelev, Alexandre Zagoskin, Alexander G. Balanov, *Chaos* **35**, 093135 (2025)
- “Super-sensing: 100-Fold Enhancement in THz Time-Domain Spectroscopy Contrast via Superoscillating Waveform Shaping”
Peisong Peng^{*}, Dustin Lindberg, **Gerard McCaul**, Denys I. Bondar, Diyar Talbayev, *APL Photonics* **10**, 086107 (2025)
- “Dynamical Learning and Quantum Memory with Non-Hermitian Many-Body Systems”
Moein N. Ivaki^{*}, Austin J. Szuminsky, Achilleas Lazarides, Alexandre Zagoskin, **Gerard McCaul**, Tapio Ala-Nissila, arXiv:2506.07676 (2025)
- “Quantum Dynamical Emulation of Imaginary Time Evolution”
Jacob Leamer, Alicia Magann, Denys I. Bondar, **Gerard McCaul**^{*}, arXiv:2403.03350 (2024).
Introduces a constructive method for simulating non-unitary dynamics via weighted ensembles of unitary evolutions. The resulting ITQDE protocol enables spectral estimation and thermal state preparation on existing quantum processors. I conceived the method, derived the formalism, and wrote the initial numerical implementation.
- “Extracting Correlation Length in Mott Insulators by Strong-field Driving”
Abdallah AlShafey, Xu-Yan Jia, Yuan-Ming Lu, Shou-Shu Gong, **Gerard McCaul**, Denys Bondar, Mohit Randeria, Takashi Oka, Alexandra S. Landsman, *J. Opt. Soc. Am. B* **41**, B26 (2024)
- “Superoscillations Deliver Superspectroscopy”
Gerard McCaul^{*,†}, Peisong Peng[†], Monica Ortiz Martinez, Dustin Lindberg, Diyar Talbayev, Denys I. Bondar, *Phys. Rev. Lett.* **131**, 153803 (2023)
- “The Waveoperator Representation of Quantum and Classical Dynamics”
Gerard McCaul^{*}, Dmitry V. Zhdanov, Denys I. Bondar, *Phys. Rev. A* **108**, 052208 (2023).
Introduces a purified square-root representation of the density matrix providing a singularity-free procedure for taking the classical limit of both real- and imaginary-time quantum dynamics.
- “Ultrafast Laser-driven Dynamics in Metal–Insulator Interface”
Abdallah AlShafey^{*,†}, **Gerard McCaul**[†], Yuan-Ming Lu, Xu-Yan Jia, Shou-Shu Gong, Zachariah Addison, Denys I. Bondar, Mohit Randeria, Alexandra S. Landsman, *Phys. Rev. B* **108**, 144434 (2023)
- “Spectral Gaps via Imaginary Time”
Jacob Leamer^{*}, Alicia Magann, Andrew Baczewski, **Gerard McCaul**, Denys I. Bondar, arXiv:2303.02124 (2023)
- “Dynamical Generation of Epsilon-Near-Zero Behaviour via Tracking and Feedback Control”
Jacob Masur^{*}, Denys I. Bondar, **Gerard McCaul**, arXiv:2301.12069 (2023)
- “Towards Single Atom Computing via High Harmonic Generation”
Gerard McCaul^{*}, Kurt Jacobs, Denys I. Bondar, *Eur. Phys. J. Plus* **138**: 123 (2023) — **Journal Highlight**
- “Free to Harmonic Unitary Transformations in Quantum and Koopman Dynamics”
Gerard McCaul^{*}, Denys I. Bondar, *J. Phys. A* **55**, 434003 (2022)
- “Non-Uniqueness of Driving Fields Generating Non-Linear Optical Response”
Gerard McCaul^{*}, Alexander F. King, Denys I. Bondar, *Annalen der Physik*, 2100523 (2022)

- “Optical Distinguishability of Mott Insulators in Time vs Frequency Domain”
Jacob Masur*, Denys I. Bondar, **Gerard McCaul**, *Phys. Rev. A* **106**, 013110 (2022)
- “Sequential Optical Response Suppression for Chemical Mixture Characterization”
Alicia B. Magann*,[†], **Gerard McCaul**[†], Herschel A. Rabitz, Denys I. Bondar, *Quantum* **6**, 626 (2022)
- “Optical Indistinguishability via Twinning Fields”
Gerard McCaul*, Alexander F. King, Denys I. Bondar, *Phys. Rev. Lett.* **127**, 113201 (2021)
- “Diffusion Fails to Make a Stink”
Gerard McCaul*, Andreas Mershin, Denys I. Bondar, *Physics of Fluids* **33**, 031801 (2021) — **Invited, Featured Article**
- “Decoherence-free Entropic Gravity: Model and Experimental Tests”
Alex J. Schimmoller*, **Gerard McCaul**, Hartmut Abele, Denys I. Bondar, *Phys. Rev. Research* **3**, 033065 (2021)
- “How to Win Friends and Influence Functionals: Deducing Stochasticity From Deterministic Dynamics”
Gerard McCaul*, Denys I. Bondar, *Eur. Phys. J. Spec. Top.* **230**, 733 (2021)
- “Fast Computation of Dissipative Quantum Systems with Ensemble Rank Truncation”
Gerard McCaul*, Kurt Jacobs, Denys I. Bondar, *Phys. Rev. Research* **3**, 013017 (2021)
- “Driven Imposters: Controlling Expectations in Many-Body Systems”
Gerard McCaul*, Christopher Orthodoxou, Kurt Jacobs, George H. Booth, Denys I. Bondar, *Phys. Rev. Lett.* **124**, 183201 (2020) — **Editor’s Suggestion**, *Physics viewpoint*
- “Controlling Arbitrary Observables in Correlated Many-body Systems”
Gerard McCaul*, Christopher Orthodoxou, Kurt Jacobs, George H. Booth, Denys I. Bondar, *Phys. Rev. A* **101**, 053408 (2020) — **Editor’s Suggestion**
- “Entropy Non-Conservation for Hamiltonian Dynamical Systems”
Gerard McCaul*, Alexander Pechen, Denys I. Bondar, *Phys. Rev. E* **99**, 062121 (2019)
- “Driving Spin-boson Models From Equilibrium Using Exact Quantum Dynamics”
Gerard McCaul*, C. D. Lorenz, L. Kantorovich, *Phys. Rev. B* **97**, 224310 (2018)
- “Partition-free Approach to Open Quantum Systems in Harmonic Environments: An Exact Stochastic Liouville Equation”
Gerard McCaul*, C. D. Lorenz, L. Kantorovich, *Phys. Rev. B* **95**, 125124 (2017)

SELECTED
INVITED
PRESENTATIONS

- **NEXT-AI 2026**, Loughborough — April 2026 — *Dynamics as Computation*
- **Chapman Institute of Quantum Studies** — October 2025 — *Free Snacks in Quantum Complexity*
- **Aalto Summer School on New Directions in Quantum and Quantum Reservoir Computing** — August 2025 — *Computation as Dynamics* and *Free Snacks in Quantum Complexity*
- **Kharkiv Quantum Seminar** — February 2025 — *Quantum Dynamical Emulation*
- **Frontiers of Quantum and Mesoscopic Thermodynamics** — July 2024 — *Thermal States via Quantum Dynamical Emulation* (also invited for 2026 session)
- **Superoscillations in Cetraro III** — June 2024 — *Superoscillations Deliver Superspectroscopy*
- **IQST Early Career Researchers Retreat** — May 2024 — *Quantum Dynamical Emulation*
- **University of Maryland, Deffner Group** — December 2023 — *Thermalisation in the Wave-operator Representation*
- **Ohio State University Institute for Optical Science** — November 2023 — *Malleable Light: From Single Atom Computing to Superspectroscopy*
- **Sandia National Laboratory** — August 2023 — *Imaginary Time in Quantum Circuits*

AWARDS & RECOGNITION

- Five papers selected as [journal highlights](#), featured articles, or editor’s suggestions. Coverage in *Physics*, *Physics World*, *Chemistry World*, *Nature Materials*, and *Quanta*.
- (2026) 2nd place FQxI ‘How Quantum Is Life?’ essay contest (\$7,500) — essay: ‘[Spanspermia: Does Life Come from Outer Hilbert Space?](#)’
- (2024) [Inaugural Tulane School of Science and Engineering full-time researcher award](#) for significant research accomplishments in quantum control and optics.
- (2023) Runner-up, University of Maryland [Quantum Thermodynamics Hub](#) short story contest (\$1,000) — entry: ‘[Doctor Up and Mister Down](#)’
- (2022) IOP Trusted Reviewer.

RESEARCH FUNDING

- Co-PI, Army Research Office grant #W911NF-23-1-0288, “Malleable Light: Programming Optical Properties via Feedback and Superoscillations” — June 2023 (\$469,382)
- Research collaboration, Sandia National Laboratory (DOE funding) — August 2023 (~\$2,000)
- Research visit, Ulm Institute for Complex Quantum Systems — May–June 2023 (~\$2,500)
- Research visit, ELTE University, Budapest — January 2025
- Visiting Student Research Collaborator (VSRC), Princeton University — January–March 2018 (~£4,000)
- Tulane postdoctoral travel awards $\times 4$ — 2019–2023 (\$2,500)
- Tulane supporting impactful publications award — March 2023 (\$2,800)
- CANES workshop organisation funding (with Overleaf sponsorship) — June 2017 (~£2,000)

TEACHING & SUPERVISION

- Supervised multiple postgraduate, undergraduate, and school students; five published papers resulting from this supervision.
- Principal supervisor, MSc thesis: “Numerical Investigation of Minimal Quantum Reservoirs” (Roland Weiszmueller, ELTE University, June 2025).
- Lecturer, General Physics II (calculus-based), Tulane University — January–May 2022. Designed course covering electrostatics, circuits, EM waves, optics, and special relativity. Mentored one student to a successful Goldwater Fellowship application.
- Teaching assistant, Mathematics for Physicists II & III, King’s College London — 2015–2018. Delivered full lectures in the final year.
- IB and IGCSE teacher (physics and mechanics), Qidong Middle School, China — 2013–2014.
- Private tutoring in advanced physics (variational principles, GR, QFT, quantum information) — 2016–present.

CONFERENCES ORGANISED

- [Micro March Meeting](#), Denver — March 2020. Organised at short notice after APS March Meeting COVID cancellation; ~30 attendees, 12 talks.
- [Workshop on Localisation in Quantum Systems](#), London — June 2017. Organised with CDT colleagues; 4 speakers, ~40 attendees.

REFEREEING

25+ reviews across Nature Communications, Physical Review Letters/A/B/Research, New Journal of Physics, European Journal of Physics B/Plus, Journal of Physics A, Entropy.

OUTREACH & ENGAGEMENT

- *Little Learning Machine* — Substack, ongoing. Public writing on the physics of computation, learning, and representation, developed in parallel with the formal research programme.
- Pint of Science talk: “[Linear Algebra Ruined Your Life](#)” (on the ubiquity of linear algebra as a framework for physical and computational dynamics) — May 2025.
- Lecture series on vector spaces and their applications, Benjamin Franklin High School (AP Mathematics), New Orleans — December 2022–January 2023.
- Mentor, US Army Educational Outreach Program (AEOP) Apprenticeships — 2020–2023.
- [GNOSEF](#) judge, Tulane University — February 2020–2023.
- [FIRST Lego League](#) judge, Jesuit High School, New Orleans — January 2020–2023.
- Physics of Utopia exhibition, London — August 2016.

SKILLS & OTHER

- Python, Maple, MATLAB, Mathematica, $\text{\textit{TikZ}}$, $\text{\textit{L}^A\text{\textit{T}}_E\text{\textit{X}}$.
- Hanyu Shuiping Kaoshi (HSK) 2 Mandarin qualification.
- Dreadful guitarist.